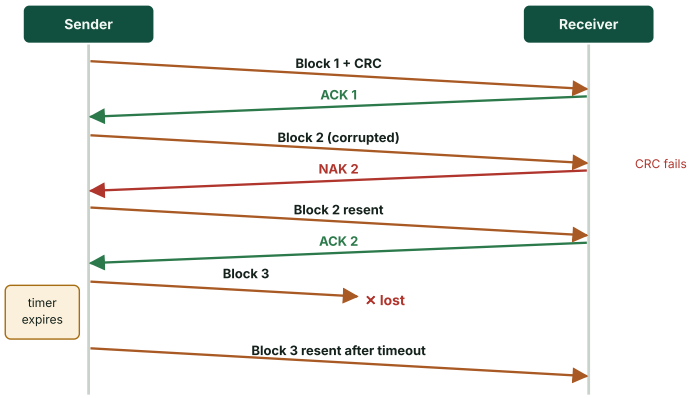
Data transmission and error control

IS12 FACT FILE

Specification: students should be able to

- Compare transmission media
- Distinguish bandwidth and bit rate and explain broadband
- Compare serial and parallel, simplex, half-duplex and full-duplex, and synchronous and asynchronous transmission
- Describe error detection (parity, checksum, CRC) and correction (ARQ)

Key words: bandwidth, bit rate, broadband, serial, parallel, simplex, duplex, synchronous, asynchronous, parity, checksum, CRC, ARQ



Automatic repeat request: block 1 is acknowledged, block 2 arrives with an error and is negatively acknowledged then resent, and block 3 is lost so a timeout triggers retransmission

Media and capacity

Twisted-pair copper is cheap and common in LANs but suffers interference and is limited to about 100 metres per run. Fibre-optic cable carries light, giving very high bandwidth over long distances, immunity to electrical interference and better security, but costs more to install. Wireless media allow mobility but are prone to interference and interception. Bandwidth is the capacity of a channel; bit rate is the number of bits actually sent each second. Transfer time equals the number of bits divided by the bit rate.

Modes of transmission

Serial transmission sends bits one after another along a single channel and works reliably over long distances. Parallel sends several bits at once but suffers skew and crosstalk, so it is used only over short distances. Simplex sends in one direction, half-duplex in both directions but not at once, and full-duplex in both directions simultaneously. Synchronous transmission uses a shared clock and sends continuous blocks; asynchronous transmission frames each character with start and stop bits.

Error detection and correction

A parity bit makes the number of 1s even or odd and detects any single-bit error but not an even number of flipped bits. A checksum is calculated from the data block and recalculated by the receiver. A cyclic redundancy check divides the data by a fixed polynomial and checks the remainder, detecting burst errors reliably. Automatic repeat request corrects errors by retransmission: the receiver sends acknowledgements, and the sender resends any block that is negatively acknowledged or whose acknowledgement does not arrive in time.

Key terms from the mark schemes

Bandwidth
The capacity of a communication channel (range of frequencies), which determines the maximum rate at which data can be transmitted.

Bit rate
The number of bits transmitted per second over a communication channel.

Full-duplex
A transmission mode in which data can travel in both directions at the same time.

Asynchronous transmission
Transmission without a shared clock, where each character is framed by start and stop bits.

Parity bit
An extra bit added to a group of bits so that the total number of 1s is even (even parity) or odd (odd parity), used to detect errors.

Automatic repeat request (ARQ)
An error-correction method in which the receiver acknowledges each block and the sender retransmits any block that is negatively acknowledged or times out.

Cyclic redundancy check (CRC)
An error-detection method in which a remainder calculated by dividing the data by an agreed polynomial is sent and checked by the receiver.

Exam focus

For calculations, write the formula, convert bytes to bits (multiply by 8), keep units consistent and give units in the answer; method marks are available even if the final answer is wrong. For error control, a full description of ARQ mentions acknowledgement, timeout and retransmission. When asked about a limitation of parity, say it cannot detect an even number of bit errors and cannot locate the error. Justify media choices in context: distance, bandwidth, interference, cost and security.

? Check yourself

1. Distinguish between bandwidth and bit rate. [2]
2. Calculate how long it would take to transfer a 250 MB file over a 20 Mbps connection. Show your working. [3]
3. Add an even parity bit to 1101001 and explain one limitation of parity checking. [3]

Answers: 1. Bandwidth is the range of frequencies a channel can carry, which determines its maximum data capacity (1) / bit rate is the number of bits actually transmitted per second (e.g. Mbps), which may be lower due to congestion, distance or interference (1). 2. Convert to bits: 250 MB × 8 = 2000 Mb (megabits) (1) / time = 2000 ÷ 20 (1) / = 100 seconds (1 minute 40 seconds) (1). Accept 104.9 s (≈105 s) if 1 MB = 1024 × 1024 bytes and 20 Mbps = 20 000 000 bps is stated. 3. 1101001 contains four 1s, which is already even (1) / so the even parity bit is 0: 11010010 (accept 01101001 if parity bit placed first) (1) / Limitation, any one: cannot detect an even number of bit errors, e.g. two bits flipped (1) / cannot identify which bit is wrong, so cannot correct the error (1).